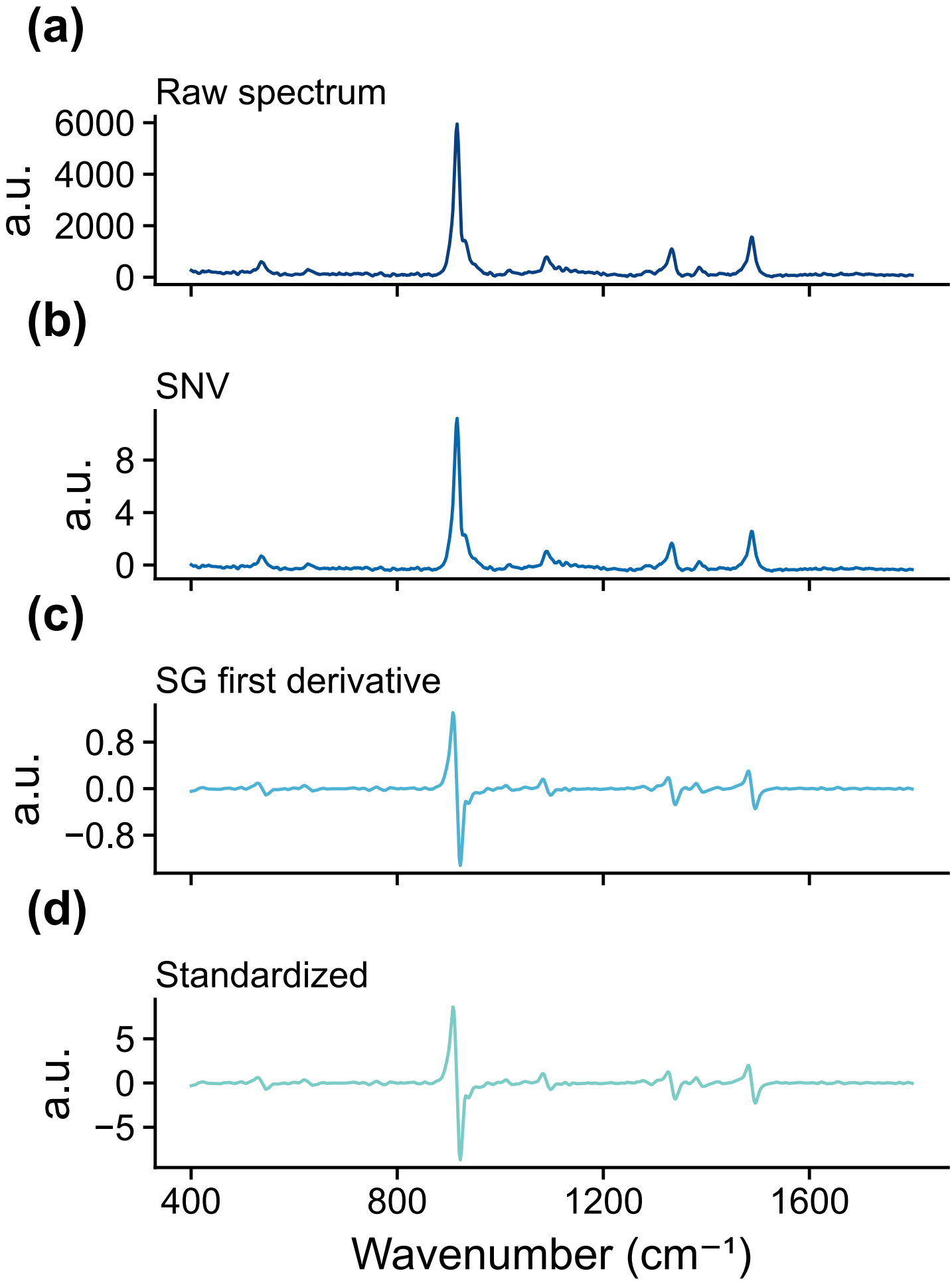
(a)
Raw spectrum
a.u.
6000
4000
2000
0
(b)
SNV
a.u.
8
4
0
(c)
SG first derivative
a.u.
0.8
0.0
−0.8
(d)
Standardized
a.u.
5
0
−5
400
800
1200
1600
Wavenumber (cm⁻¹)